

This assignment will not be graded and is only for practice.

Level 1

1) Choose the correct statements about $f(x, y) = x^3 + xy + y^4 - 4$.

1 point

- ☐ $(0, 0)$ is a critical point of f .
- ☐ $(0, 0)$ is the only critical point of f in $\mathbb{R}^2$.
- ☐ $f_{xx} = 6x$
- ☐ $f_{yy} = 12x^2$
- ☐ $f_{xy} = 1$
- ☐ $f_{yx} = 0$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 0)$ is a critical point of f .

$$f_{xx} = 6x$$

$$f_{xy} = 1$$

2) If $f(x, y) = \tan^{-1}(x/y)$, then which of the following options are correct.

1 point

- ☐ $f_x = \frac{y}{(x^2+y^2)}$
- ☐ $f_{yy} = \frac{2xy}{(x^2+y^2)^2}$
- ☐ $f_{yy} = \frac{-2xy}{(x^2+y^2)^2}$
- ☐ $f_{yx} = \frac{x^2-y^2}{(x^2+y^2)^2}$
- ☐ $f_{yx} = \frac{y^2-x^2}{(x^2+y^2)^2}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$f_x = \frac{y}{(x^2+y^2)}$$

$$f_{yy} = \frac{2xy}{(x^2+y^2)^2}$$

$$f_{yx} = \frac{x^2-y^2}{(x^2+y^2)^2}$$

3) If $f(x, y) = x^2 - 3xy^3$, then the determinant of the Hessian matrix $Hf(x, y)$ is **1 point**

- ☐ $36xy + 81y^4$
- ☐ 0
- ☐ $-36xy - 81y^4$
- ☐ $36xy - 81y^4$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$-36xy - 81y^4$$

Level 2

4) Choose the correct statements about $f(x, y) = x^2 + y^4$.

1 point

- ☐ $(0, 0)$ is a critical point of f .
- ☐ Rank of the Hessian matrix $Hf(0, 0)$ is 2.
- ☐ The determinant of the Hessian matrix $Hf(0, 0)$ is 0.
- ☐ The determinant of the Hessian matrix $Hf(0, 0)$ is 1.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 0)$ is a critical point of f .

The determinant of the Hessian matrix $Hf(0, 0)$ is 0.

5) Number of critical points of the function $f(x, y) = x^3 + y^3 - 3x - 3y$ is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 4

1 point

6) For which of these functions are the mixed partial derivatives the same at $(0, 0)$.

1 point

- ☐ $f(x, y) = 3x^2 + 7y^3 - 2x^2y^3$.
- ☐ $f(x, y) = \begin{cases} 1 & y = 0 \\ 0 & y \neq 0 \end{cases}$.
- ☐ $f(x, y) = |x|(y^2 + 1)$.
- ☐ $f(x, y) = ax + by$, where a and b are two fixed real numbers.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$f(x, y) = 3x^2 + 7y^3 - 2x^2y^3.$$

$$f(x, y) = ax + by, \text{ where } a \text{ and } b \text{ are two fixed real numbers.}$$

Your score is: 0/6